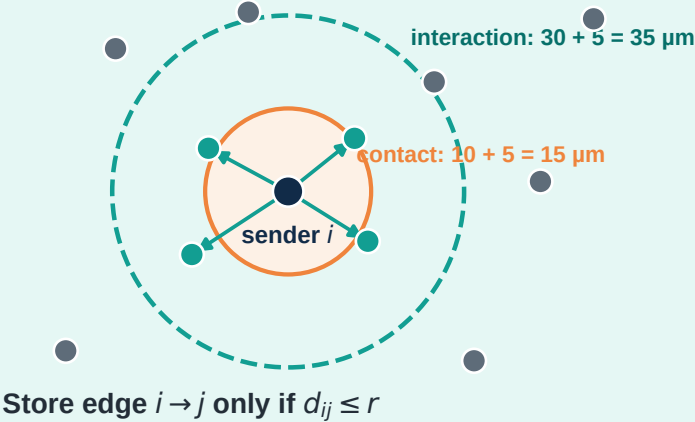


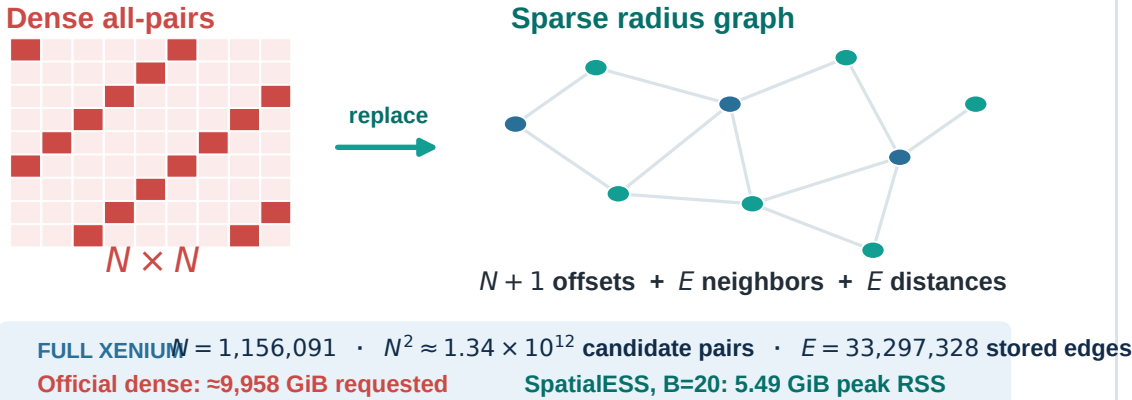
How SpatialESS stores spatially plausible communication

A sparse radius graph and CSR neighbor directory replace the dense all-pairs distance matrix—without changing the official probability formula.

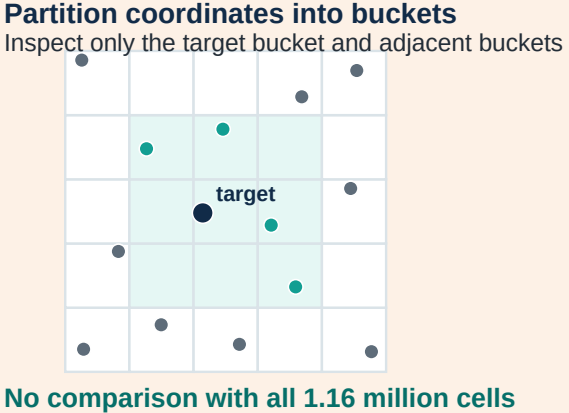
A Radius-limited communication



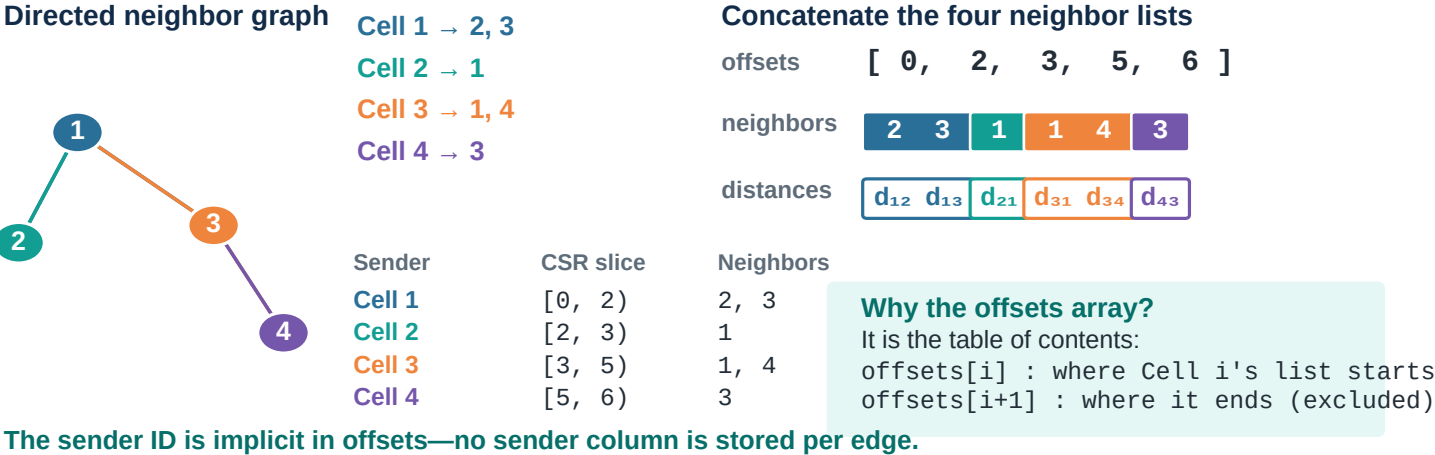
B Store local edges, not all pairs



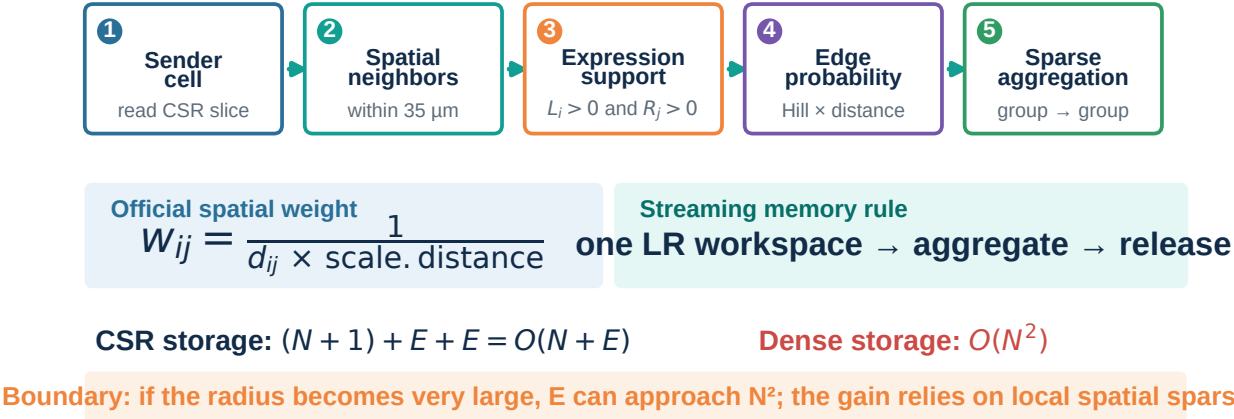
C Spatial hashing



D CSR is a compact neighbor directory



E Stream communication over CSR



Take-home: neighbors stores who can communicate; distances stores how far apart they are; offsets tells C++ where each sender's neighbor list begins and ends.

SpatialESS v3-compatible benchmark: 1,156,091 cells · 33,297,328 directed edges · 627 LR · 20 permutations · 27.19 min · 5.49 GiB peak RSS.